

Mücahid Enes Deniz

Software Engineer

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SUMMARY

Outcome-driven developer with hands-on experience building production systems in computer vision, industrial automation, and machine learning. Proven ability to design and deploy real-world solutions using C++, Python, and SQL across manufacturing environments. Motivated to solve complex problems and scale high-impact technologies.

EDUCATION

- Bahçeşehir University** Istanbul, TR
B.Sc., Computer Engineering (Full Scholarship) | GPA: 3.07/4.00 Sep 2020 - Jun 2025
- Bornova Anatolian High School** Izmir, TR
Science & Math | GPA: 93.3/100 Sep 2016 - Jun 2020

WORK EXPERIENCE

- Kastamonu Entegre (Hayat Holding)** Istanbul, TR
Software Engineer (Full-time) Jul 2025 - Present
 - Adaptive Optical Quality Control (Patented):** Engineered a multi-site hybrid C++/Python vision system deployed across 3 factories. Designed a real-time Redis IPC architecture linking a Qt6 GUI with Python automation. Integrated YOLOv8 via ONNX Runtime for CPU inference and Snap7 for Siemens S7 PLC communication. Implemented an automated recovery orchestrator, delivering zero-downtime operations and reducing CAPEX by 88%.
 - AI Line Inspection:** Developed a real-time industrial quality control system utilizing a 5-stage asynchronous producer-consumer pipeline. Built a zero data-loss architecture with RAM-bounded queues to ensure 100% automated coverage on high-speed lines. Orchestrated multi-threaded YOLOv11 CPU inference and high-speed image stitching with Basler cameras.
 - Dimensional Control:** Implemented sub-millimeter geometric squareness measurement on moving sheets using dual Basler GigE line-scan cameras; applies an OpenCV pipeline (morphological filtering, Canny, Hough Transforms) with a per-sheet dynamic mm/px ratio derived from live PLC data to compute both diagonals and their difference. Logged to Redis and SQL for full traceability.
 - Weighbridge (3D Volumetric Scanning):** Built 4x SICK LiDAR fusion reconstructing real-time 3D point clouds of moving trucks for volume/tonnage estimation; achieved 95% precision via motion compensation. Delivered operator dashboards via Flask API with PostgreSQL analytics.*TalenTree Intern (Long-term)* Jul 2024 - Jun 2025
 - Contributed to the design, implementation, deployment, and maintenance of industrial AI, automation, and computer vision systems used in production environments.
 - Participated in Agile/Scrum processes and served as Scrum Master in a Sustainable Supplier Selection project.
- Harpeon** Istanbul, TR / Remote
Mobile Application Development Intern Jun 2023 - Aug 2023
 - Built a Weather app with external API integration and prototyped an office-reservation UI using React Native and Expo.

PROJECTS

- PAC-MAN Clone (Unity, C#):** Grid-based movement with input buffering, four-state Ghost AI (State Pattern + BFS + Line-of-Sight chase trigger), flood-fill dot placement, endless respawn, and persistent high score.
- Mobile Match-3 Game (Unity, C#):** Blast-style puzzle game with iterative BFS matching, rocket power-ups, obstacle-aware gravity, top refill, and JSON-driven level design.
- Grid Survival Game (Godot, GDScript):** Grid-based movement with dynamic area expansion power-ups, shield mechanics, bomb spawning, and score tracking.
- M&M Chocolate Detector and Sorter (Python):** Built a real-time pipeline with a custom model for M&M vs. non-M&M and color classification; Arduino-driven actuators handled physical sorting.
- Education App (Java):** Android application with 10 mini-games for children with dyslexia; Firebase authentication, bilingual (TR/EN) support, and separate Learn/Play activity tracks.

SKILLS

Programming: Python, C++, C#, JavaScript, SQL **AI & Computer Vision:** YOLO, ONNX Runtime, OpenCV, ML
Tools & Frameworks: Qt, Flask, Git, Redis, PostgreSQL, Power BI **Industrial:** Basler Pylon, Siemens S7 PLC, SICK LiDAR

LANGUAGES & INTERESTS

Languages: Turkish (Native), English (Advanced), Turkish Sign Language
Interests: Esports (LoL), Salsa, Chess, Running, Taekwondo